C-HI Run Configuration Audit

Date: 2026-05-13

Summary

This audit answers the configuration-review request from the third Barry-Stas- Patrick meeting: expose the actual grid, output sizes, loaded modules, run parameters, node/rank layout, and source/configuration provenance for the current reduced-grid `C-HI` run before we move toward a full production launch.

The important conclusion is simple: job `691422.kunanyi-ohpc` is a successful 8-node reduced-grid continuation, not the final full-grid production configuration. It is excellent as a restart/output/observable rehearsal, but it should not be mistaken for the final grid that Stas and Patrick need to review before an 80-node-style production run.

Run Identity

Item	Value
Source run	`691422.kunanyi-ohpc` (`c_hi_8mini`)
Case	`C-HI`, Stewart cluster environment
Variant	particles enabled
Restart mode	`-h5restart 7` from `data.0007.dbl.h5`
Run args	`-i pluto.ini -e environment.ini -maxtime 14.5 -h5restart 7`
Node/rank layout	`select=8:ncpus=28:mpiprocs=28`, 224 MPI ranks
Host nodes	`n063`, `n064`, `n072`, `n135`, `n169`, `n180`, `n212`, `n216`
Start time	`10.278362 Myr`
Final time	`33.374180 Myr`
Final restart checkpoint	`data.0012.dbl.h5`

Environment

The live run used the supervisor-confirmed radially averaged/tabulated Stewart cluster environment:

Item	Value
Environment kind	tabulated radial profile
Environment source	`env/extracted/environments/tabulated/tabulated_cosmo_`
Runtime file	`tabulated_cosmo_002003.csv`
`environment.ini` external cube fields	deliberately unused for this tabulated run
Density profile define	`ENV_TABULATED`

This follows Stas' instruction that science runs should use the radially averaged environments rather than the full 3D profiles.

Grid

The generated `grid.out` for this run is:

Axis	Domain	Cells	Central `dx`	`dx` near 200 kpc	Edge `dx`
X1	`-200..200 kpc`	`404`	`0.05 kpc`	`11.08 kpc`	`11.08 kpc`
X2	`-200..200 kpc`	`404`	`0.05 kpc`	`11.08 kpc`	`11.08 kpc`
X3	`-250..250 kpc`	`452`	`0.05 kpc`	`8.39 kpc`	`10.44 kpc`

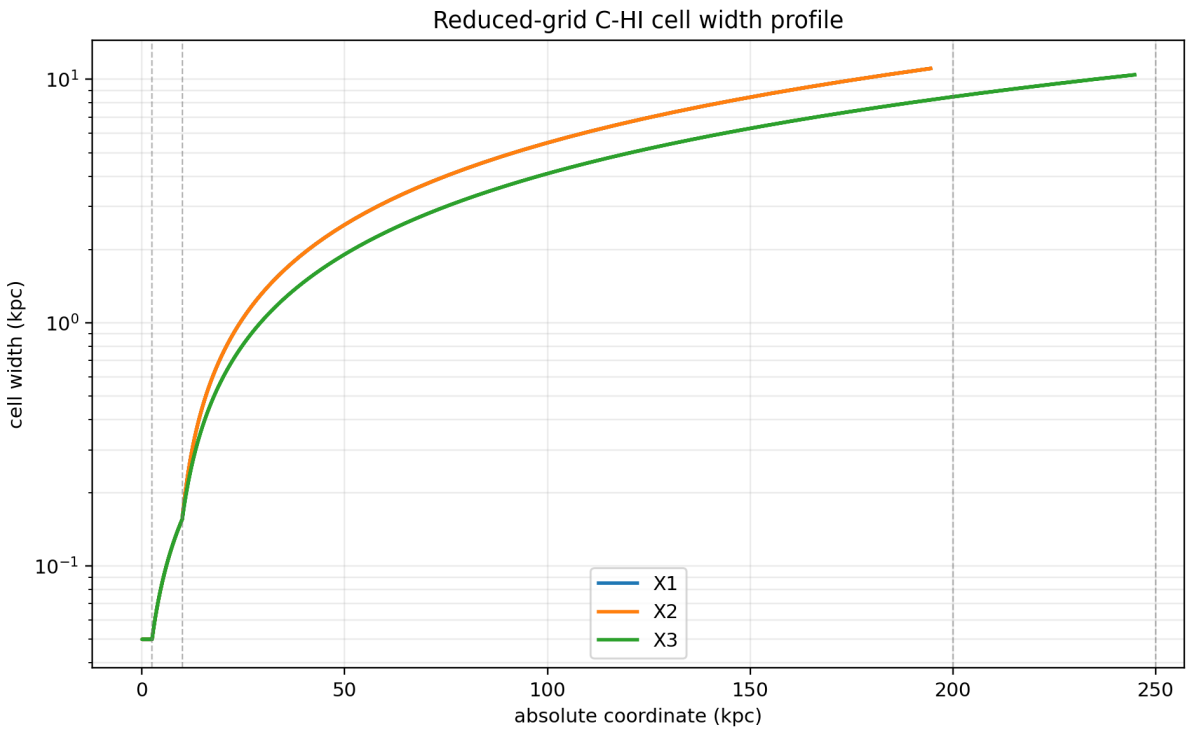


Figure 1: Reduced-grid C-HI cell widths

The run manifest names this grid `kunanyi_single_node_250` and describes it as a reduced-grid shakedown: X1/X2 retain `+/-200 kpc`, X3 extends to `+/-250 kpc`, and the central jet/inlet region keeps `0.05 kpc` resolution. The outer stretch counts were deliberately reduced so the particle-enabled problem could be built, restarted, and continued cheaply.

That means the present grid is not the candidate final production grid. The candidate production grid remains `stewart_stretched_250`, with much larger outer stretch counts and the X3 extension Patrick and Stas requested. That full-grid render should be the next object of collaborator-agent review before the first 48-hour production job.

Jet And Output Parameters

Parameter	Value
`JET_PWR`	`1e45 erg/s`
`JET_SPD`	`0.1 c`
`JET_CHI`	`1.0`
`JET_OA_PRIMARY` / `JET_OA_SECONDARY`	`15 deg` / `15 deg`
`JET_INITIAL_RADIUS`	`0.1 kpc`

Parameter	Value
JET_INJECTION_HEIGHT	0.75 kpc
JET_END_TIME / JET_ACTIVE_TIME	50 Myr / 50 Myr
JET_RELATIVISTIC_DENSITY	NO
Nparticles	-1 16.0
Particle injection frequency	0.01 Myr

Output	Cadence	Files after 691422	Size
flt.h5	1 Myr	40 (data.0000-0039.flt.h5)	3.55 GB each; 141.90 GB total
dbl.h5	5 Myr plus walltime stops	13 (data.0000-0012.dbl.h5)	5.32 GB each; 69.13 GB total
particles.dbl	0.01 Myr	3344 (particles.0000-3343.dbl)	up to 12.83 MB; 21.44 GB total

The output footprint after the 8-node job was about 217G on scratch.

Runtime Stack

The PBS job loaded HDF5/1.14.0-gompi-2023a, which pulled the expected compiler/MPI stack:

Component	Version/path
HDF5	HDF5/1.14.0-gompi-2023a
HDF5 directory	/share/apps/software/HDF5/1.14.0-gompi-2023a
OpenMPI	OpenMPI/4.1.5-GCC-12.3.0
mpicc	/share/apps/software/OpenMPI/4.1.5-GCC-12.3.0/bin/mpicc
mpirun	/share/apps/software/OpenMPI/4.1.5-GCC-12.3.0/bin/mpirun
Compiler stack	GCC/12.3.0, GCCcore/12.3.0
MPI transport stack	UCX/1.14.1, PMIx/4.2.4, UCC/1.2.0, libfabric/1.18.0

Provenance

For job 691422, the available source/configuration evidence is:

Item	Evidence
case_manifest.json SHA-256	eaa7aa84c428aef13ae24d0dc5317589c17b160ee3426c5b81c7c
pluto.ini SHA-256	b9c0a7313059c9f1b31ba4846c502ecb8597a274d3b612ad8bf3c
environment.ini SHA-256	3b49690db1185c70742400925fb19f3f51abead2967f0e46a732b

The current source_revisions.txt did not record PLUTO or jet-setup commit SHAs, only the scratch paths and config-file hashes. I have updated the production PBS template so future jobs explicitly record git root, branch, head SHA, status, missing-git state, and the PLUTO binary SHA-256. That fix should be staged before the next production-template submission.

The follow-up particle-HDF5 rebuild for the through-33-Myr diagnostics completed after being submitted with a temporary scheduler-fit 16gb memory request, but PBS reported roughly 43 GB memory use. I have updated that rerun wrapper to request 48gb for future submissions.

Gate Status

This audit supports three decisions:

- The reduced-grid lane is operationally healthy: restart, HDF5 output, particle output, email notification, and multi-node MPI all worked.
- It is not yet a full production science grid: the outer-cell widths in this reduced run are deliberately coarse and should not be used as the final configuration.
- The next production-facing gate should be a full-grid configuration render and review, followed by one bounded kunanyi production-template run only after Patrick/Stas or their repo agents confirm the final grid/config.

Generated Artifacts

Supporting files are in:

```
docs/science/assets/c_hi_run_configuration_audit_2026-05-13/
```

Key files:

- run_config_summary.json
- grid_resolution_summary.csv
- output_file_size_summary.csv

- `flt_output_times.csv`
- `dbl_output_times.csv`
- `c_hi_reduced_grid_cell_widths.png`